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Determinants of Child Mortality: Evidence from the 2022 Uganda DHS

1. INTRODUCTION

Child mortality, measured by the under-five mortality rate (U5MR), is the probability that a child born in a given year will die before reaching five years. According to the United Nations Department of Economic and Social Affairs (UN DESA), reducing child mortality to less than 25 deaths per 1,000 live births by 2030 remains a global challenge (UN DESA 2025). The recent Sustainable Development Goals (SDGs) report reveals that U5MR only decreased from 44 deaths per 1,000 live births in 2015 to 37 deaths in 2023 (UN DESA 2025). This implies that millions of children across the world die before their fifth birthday.

Poor and middle income countries contributes the largest share of child deaths. World Health Organization (WHO) reported that 58 percent of child deaths, that is about three million deaths in 2024, occurred in sub-Saharan Africa (WHO 2026). In Uganda, which is the focus of this analysis, the Bureau of Statistics (UBOS) revealed that U5MR only reduced to 46 deaths per 1,000 live births in 2024 (UBOS 2025). This rate is almost twice the 2030 target of less than 25 deaths per 1,000 live births. This study examines the factors associated with child mortality in Uganda. This is an important quest because understanding the predictors of child deaths is essential for designing and implementing interventions to improve child health outcomes.

The dataset analysed consisted of 1434 survey observations. It was obtained from the Kid's Recode file of the 2022 Uganda Demographic and Health Survey (UDHS). Statistical software R and Stata were used to extract, clean, recode, and analyse the data. The results reveal the total number of children born and birth interval as main predictors of child survival in Uganda. Giving birth to another child was associated with 24 percent higher odds of child death. Children born after an interval of 24–47 months experienced lower odds of mortality compared to those born after an interval of less than 24 months.

Covariates such as maternal education, wealth, place of residence, maternal age, and source of water were not statistically significant predictors of child mortality.

These findings point to a need to strengthen reproductive health and family planning in Uganda. Particularly, policy interventions that promote optimal birth spacing of at least more than 24 months and minimal number of children are essential for child survival. In addition, expanding maternal access to family planning services and overall child healthcare in rural areas may positively influence child survival in Uganda.

1.1. Objective

- This analysis utilized the recent 2022 Demographic and Health Survey (UDHS) data to assess the determinants of child mortality in Uganda.

1.2. Research Question

- Does child, maternal, and socioeconomic factors predict child survival in Uganda?

2. DATA AND METHODOLOGY

2.1. Sample and Data Source

This study utilized the Kid's Recode (KR) data of the 2022 Uganda Demographic and Health Survey (UDHS). UDHS is a nationally representative survey of mothers between 15 and 49 years, conducted every five years by the Uganda Bureau of Statistics (UBOS). UBOS employed a stratified two-stage cluster sampling design, with enumeration areas serving as primary sampling units (PSUs) at the first stage and households selected within PSUs at the second stage. The sampling frame covered 13 sub-regions, capturing both rural and urban populations. The KR file consisted of 1,434 children below the age of five years.

2.2. Outcome Variable

The main outcome of interest was the child survival status, coded as a binary indicator where 1 denoted child death and 0 denoted child survival.

2.3. Explanatory Variables

The explanatory variables were selected based on Mosley and Chen (1984) framework for the analysis of child mortality. The framework provides a list of child, maternal, and household factors relevant for the analysis of child survival predictors. The child-level variables selected include: Child sex (female; male), age, and birth order. The birth order variable was collapsed into three categories (1st; 2nd–3rd; and 4th and above) for simplicity purpose. Maternal variables included: Age, birth interval, total children born, and education. The birth interval variable was also collapsed into three categories (<24 months; 24–47 months; and 48+). Likewise, maternal education was also collapsed into no education, primary, secondary or higher. Household variables comprised of wealth index (poorest; poorer; middle; richer; richest), place of residence (rural; urban), region, and source of drinking water. The sources of drinking water were recoded into improved and unimproved categories. Water sources including piped water, boreholes, protected wells and springs, rainwater, tanker trucks, bottled and sacket water were classified as improved, while unprotected wells and springs, surface water, and other unspecified sources were classified as unimproved.

2.4. Statistical Analysis

The statistical software R and Stata were used for the analysis. R was used for variable extraction and descriptive data analysis. On the other hand, Stata was used for data cleaning and logistic regression analysis.

First, child mortality rates per 1,000 live births were computed by key child, maternal, and household factors, with 95% confidence intervals reported for each estimate. Second, bivariate analysis was conducted to examine the associations between each explanatory variable and child mortality. The analysis applied survey-adjusted pearson's chi-square tests with Rao-Scott correction for categorical variables, and t-tests for the continuous variables. Finally, a multivariate survey-weighted logistic regression was conducted to estimate the determinants of child mortality. The results are reported as adjusted odds ratios (AOR) with 95% confidence intervals. All the regression analyses accounted for

the survey design through PSU clustering, stratification, and probability weights, and statistical significance was assessed at the conventional 5% threshold.

3. RESULTS

3.1. Descriptive Analysis

Table I presents a sample of 1,434 children, with a nearly balanced gender distribution. Male children accounted for 50.2% of the sample, while female children accounted for 49.8%. Nearly half of the children (45%) were of fourth birth order or higher, while 34.6% were second or third borns. Only 20.4% of the children were first births.

The average preceding birth interval was 28.3 months (SD = 22.0), implying that the next child was born after more than two years. The average maternal age was 28.7 years (SD = 6.8), and the number of children was 3.9. This reflects a relatively high fertility patterns among Ugandan mothers.

The majority of the children were born to mothers with a primary level of education (59.9%) and secondary education (27.3%). A small proportion of mothers (4.1%) had a higher level of education. Those born to mothers with no formal education were only 8.7%.

Table II reveals a relatively even distribution of households across wealth categories and reliance on borehole water. A larger proportion of households (24.8%) belonged to the poorest wealth index. Households belonging to the other wealth indexes are fairly comparable. About 46.7% of the households rely on borehole water. Other reported sources of water were piped water to neighbors (9.2%), public taps (8.1%), protected springs (7.4%), surface water sources such as rivers and lakes (7.3%), and unprotected wells (7.0%). Relatively few households had direct access to piped water within their premises (1%) or plots (4.2%). Though the majority of the households did have access to improved water sources, others still rely on unimproved sources such as unprotected wells, springs, rivers, lakes, and ponds.

Table III reveals that the majority of children (70%) resided in rural areas and only 30% lived in urban areas. This reflects Uganda's population distribution where the majority

of the people live in rural areas. Buganda region, which included Kampala, dominated with 27.5% of the sample. It was followed by Busoga with about 11%. The remaining regional distribution range between 2.6% (Kigezi) and 11% (Busoga).

Table I: Child and Maternal Characteristics.

Characteristics	n (%), N = 1434¹
Gender of the Child	
Female	715 (49.8%)
Male	719 (50.2%)
Birth Order	
1st	286 (20.4%)
2nd-3rd	503 (34.6%)
4th+	645 (45.0%)
Birth Interval	
	28.3 (22.0)
Missing	35
Mother's Age (Years)	
	28.7 (6.8)
Mother's Education Level	
No education	135 (8.7%)
Primary	867 (59.9%)
Secondary	374 (27.3%)
Higher	58 (4.1%)
Total Children Ever Born	
	3.9 (2.4)

¹Unweighted N (weighted %); DHS sampling weights applied. Mean (SD) for all continuous variables.

Table IV presents weighted child mortality rates. The estimated U5MR was 49 deaths per 1,000 live births (95% CI: 36.0–62.0). There was a slight difference in mortality rate by child sex. Female children recorded a mortality rate of 49.4 deaths per 1,000 live

births, while male children experienced 48.7 deaths. The confidence intervals, however, overlaps which suggests that child mortality does not differ by sex of the child.

Similarly, place of residence showed only modest variation in U5MR. Rural households experienced slightly higher mortality rate (50.6 deaths per 1,000 live births) than urban households (45.3 deaths per 1,000 live births). This difference may reflect the fact that rural populations often experience poorer access to child healthcare services. However, the confidence intervals are wide and overlapping, indicating that the rural-urban gap may not be substantial.

Table II: Household socioeconomic characteristics

Characteristics	n (%), N = 1434¹
Household Wealth Index	
Poorest	382 (24.8%)
Poorer	287 (18.0%)
Middle	268 (19.3%)
Richer	245 (19.0%)
Richest	252 (18.9%)
Source of Drinking Water	
Piped into dwelling	11 (1.0%)
Piped to plot	71 (4.2%)
Piped to neighbor	122 (9.2%)
Public tap	100 (8.1%)
Borehole	707 (46.7%)
Protected well	61 (4.4%)
Unprotected well	89 (7.0%)
Protected spring	109 (7.4%)
Unprotected spring	34 (2.5%)
Rainwater	14 (1.2%)
Tanker truck	2 (0.2%)
Bicycle with jerrycans	5 (0.2%)
Surface water (river, dam, lake, pond)	98 (7.3%)
Bottled water	7 (0.4%)
Sachet water	3 (0.2%)
Missing	1

¹Unweighted n (Weighted %); DHS sampling weights applied.

Table III: Household's place of residence and region

Characteristics	n (%), N = 1434 ¹
Place of Residence	
<i>Rural</i>	1,008 (70.0%)
<i>Urban</i>	426 (30.0%)
Region	
<i>Buganda</i>	282 (27.5%)
<i>Busoga</i>	144 (10.8%)
<i>Bukedi</i>	87 (5.2%)
<i>Bugisu</i>	84 (4.4%)
<i>Teso</i>	99 (6.4%)
<i>Karamoja</i>	84 (6.2%)
<i>Lango</i>	98 (6.7%)
<i>Acholi</i>	90 (4.3%)
<i>West Nile</i>	112 (3.7%)
<i>Bunyoro</i>	111 (7.1%)
<i>Tooro</i>	104 (7.8%)
<i>Ankole</i>	92 (7.2%)
<i>Kigezi</i>	47 (2.6%)

¹Unweighted n (Weighted %); DHS sampling weights applied.

Table IV shows some socioeconomic disparities, although the pattern is not perfectly linear. Child mortality was highest among children from richer households (64.1 deaths per 1,000 live births) and poorest households (54.5 deaths). The lowest mortality rate was observed among the richest households (32.4 deaths per 1,000 live births). The absence of a smooth gradient across wealth categories may reflect sampling variability, relatively small subgroup sizes, and or the influence of other unobserved household

characteristics. Moreover, the wide and overlapping confidence intervals suggest that these subgroup differences could be due to sampling variability.

The U5MR per maternal education demonstrates a clearer pattern. Children born to mothers with a secondary education experienced substantially lower mortality (31.3 deaths per 1,000 live births) compared to children of mothers with no education (57.4 deaths) or primary education (53.3 deaths). These findings reflect the importance of education in improving healthcare utilisation. However, the U5MR for mothers with higher education appears anomalous. This is possibly because only a small proportion of women were in this category. The extremely wide confidence interval (0.0–190.0) indicates low statistical precision.

Child mortality was lowest among second- and third- born children (34.3 deaths per 1,000 live births). Children born as first and fourth or higher borns experienced higher U5MR: 52.4 deaths per 1,000 and 58.8 deaths respectively. These findings are plausible. First borns may face higher mortality due to risks associated with first pregnancies. Higher order births may reflect household resource constraints and competition among siblings for healthcare.

Table V presents child mortality rates per 1,000 live births across 13 regions in Uganda. Bunyoro recorded the highest child mortality rate of 109.6 deaths per 1,000 live births (95% CI: 46.0–173.2). Lango (65.4; 95% CI: 6.4–124.4), Acholi (60.9; 95% CI: 7.1–114.7), and Busoga (55.5; 95% CI: 16.2–94.8) also recorded high rates. Other regions' U5MR clustered near the national average of 49 deaths per 1,000 live births, including Buganda (49.7), Bukedi (50.9), West Nile (46.5), Karamoja (47.8), and Bugisu (48.4). Tooro (9.4), Kigezi (22.7), and Teso (24) recorded the lowest child mortality rates.

Table IV: Weighted child mortality rate per 1,000 live births by key subgroups

Subgroup	Category	Mortality rate	95% CI
Overall	N =1434	49.0	36.0–62.0
Gender of the Child	Female	49.4	29.9–68.9
	Male	48.7	30.7–66.6
Place of residence	Rural	50.6	35.1–66.1
	Urban	45.3	21.2–69.4
Wealth index	Poorest	54.5	29.8–79.2
	Poorer	38.6	9.2–68.0
	Middle	53.1	21.1–85.1
	Richer	64.1	27.1–101.2
	Richest	32.4	9.9–54.9
Mother's Education Level	No education	57.4	10.1–104.8
	Primary	53.3	36.6–70.0
	Secondary	31.3	9.8–52.9
	Higher	86.7	0.0–190.0
Birth order	1st	52.4	23.6–81.3
	2nd-3rd	34.3	16.7–51.9
	4th+	58.8	37.6–79.9

Weighted child mortality estimates; DHS sampling weights applied. 95% CI = 95% confidence interval. Unweighted sample size = 1434.

Table V: Regional weighted child mortality rate per 1,000 live births

Region	Child mortality rate	95% CI
Buganda	49.7	20.1–79.2
Busoga	55.5	16.2–94.8
Bukedi	50.9	1.0–100.8
Bugisu	48.4	0.0–100.0
Teso	24.0	0.0–55.9
Karamoja	47.8	5.5–90.1
Lango	65.4	6.4–124.4
Acholi	60.9	7.1–114.7
West Nile	46.5	3.4–89.7
Bunyoro	109.6	46.0–173.2
Tooro	9.4	0.0–28.1
Ankole	30.5	0.0–71.2
Kigezi	22.7	0.0–67.0

Weighted child mortality estimates; DHS sampling weights applied. 95% CI = 95% confidence interval; lower bounds truncated at zero where applicable. Wide confidence intervals reflect small regional sample sizes. Unweighted sample size = 1434.

3.2. Bivariate Analysis

Table VI illustrates bivariate associations between child mortality and child, maternal, and household characteristics. The analysis compares the distribution of characteristics between children who were alive and those who had died. The reported p-values assess whether observed differences are statistically significant.

The sample consisted predominantly of surviving children (1,364 children) as compared to dead children (70 children). Female children accounted for 50.2% of the deaths, while

male children accounted for 49.8%. This difference was not statistically significant ($p>0.05$), meaning that child survival does not differ by gender.

There was insignificant relationship between child mortality, maternal age, and birth interval. Mothers whose children died had a slightly higher mean age of 29.6 years compared to mothers of surviving children (28.6 years). The difference, however, was small and statistically insignificant ($p>0.05$). Similarly, children who died had a shorter average birth interval (26.5 months). Surviving children had 28.4 months. However, the difference was not statistically significant ($p>0.05$).

The total number of children ever born had a significant association with child mortality ($p<0.05$). On average, mothers of deceased children had more children (4.7) compared to mothers of surviving children (3.9 children). This finding suggests that having more children increases the probability of child mortality. This resonates with the descriptive findings, which revealed that most of the children belonged to poor rural households. Thus, having many children may have exacerbated income constraints and competition among siblings for nutrition and healthcare.

The relationship between maternal education and child mortality was not significant ($p>0.05$). Children born to mothers with a primary education constituted the majority of surviving (59.7%) and deceased (65.2%) children. Those born to mothers with a secondary or higher education accounted for 24.6% of child deaths relative to surviving children (31.7%). Conversely, children born to uneducated mothers had lower mortality risks compared to those whose mothers had a certain level of education.

Household wealth index did not demonstrate a statistically significant relationship with child survival ($p>0.05$). Children from the poorest households represented a slightly larger share of deaths (27.5%) compared to surviving children (24.7%). Those from the richest households accounted for smaller share of deaths (12.5%) compared to 19.2% for surviving children. These patterns align with the expectation that socioeconomic disadvantages contribute to poorer child health outcomes.

A household's place of residence also showed no significant association with child mortality ($p>0.05$). Rural households accounted for the majority of surviving (69.9%) and dead children (72.3%). For urban households, the proportion of surviving children (30.1%) was slightly higher than dead children (27.7%). The insignificant result reveals a slightly similar occurrence of child deaths among urban and rural households.

Child's birth order demonstrated some variation in mortality distribution, although the relationship was not statistically significant ($p>0.05$). Children of fourth or higher birth order accounted for more than half of child deaths (53.9%) compared to 44.5% among surviving children. In contrast, second and third born children represented a smaller proportion of deaths (24.3%) relative to surviving children (35.2%).

Finally, household's source of drinking water was not statistically associated with child mortality ($p>0.05$). Interestingly, households using improved water sources accounted for 86.9% of child deaths. This result may not necessarily mean that improved water sources leads to higher child death. In fact, most rural households relied on borehole water, protected springs, and generally, communal water sources. These sources may still provide unsafe and or poor quality water that could have caused poor child health. In addition, the equipment used for fetching and storing water could be contaminated either at the source or at home. Thus, access to a water source may not translate to access to safe water.

Table VI: Bivariate analysis of the determinants of child mortality in Uganda

Characteristics	Child survival status			p-value ²
	Overall (n = 1434) ¹	Dead (1) ¹	Alive (0) ¹	
Gender of the child				0.959
<i>Female</i>	715 (49.8%)	33 (50.2%)	682 (49.8%)	
<i>Male</i>	719 (50.2%)	35 (49.8%)	684 (50.2%)	
Mother's age (years)	28.7 (6.8)	29.6 (7.9)	28.6 (6.8)	0.361
Birth interval (months)	28.3 (22.0)	26.5 (22.3)	28.4 (22.0)	0.579
Total children born	3.9 (2.4)	4.7 (2.5)	3.9 (2.4)	0.016
Mother's education				0.589
<i>No education</i>	135 (8.7%)	9 (10.2%)	126 (8.6%)	
<i>Primary</i>	867 (59.9%)	44 (65.2%)	823 (59.7%)	
<i>Secondary+</i>	432 (31.4%)	15 (24.6%)	417 (31.7%)	
Wealth index				0.587
<i>Poorest</i>	382 (24.8%)	22 (27.5%)	360 (24.7%)	
<i>Poorer</i>	287 (18.0%)	10 (14.2%)	277 (18.2%)	
<i>Middle</i>	268 (19.3%)	14 (20.9%)	254 (19.2%)	
<i>Richer</i>	245 (19.0%)	13 (24.9%)	232 (18.7%)	
<i>Richest</i>	252 (18.9%)	9 (12.5%)	243 (19.2%)	
Place of residence				0.721
<i>Rural</i>	1,008 (70.0%)	51 (72.3%)	957 (69.9%)	
<i>Urban</i>	426 (30.0%)	17 (27.7%)	409 (30.1%)	
Birth order				0.224
<i>1st</i>	286 (20.4%)	15 (21.8%)	271 (20.3%)	
<i>2nd-3rd</i>	503 (34.6%)	17 (24.3%)	486 (35.2%)	
<i>4th+</i>	645 (45.0%)	36 (53.9%)	609 (44.5%)	
Source of drinking water				0.136
<i>Improved</i>	1,141 (78.9%)	58 (86.9%)	1,083 (78.5%)	
<i>Unimproved</i>	293 (21.1%)	10 (13.1%)	283 (21.5%)	

¹Unweighted n (weighted %); DHS sampling weights applied. Mean (SD) for continuous variables. p-values from survey-adjusted chi-square test and t-test

²Pearson's X²: Rao & Scott adjustment; Design-based t-test

□

3.3. Graphical Presentation of Child Mortality Rates

Figure I reveals that child mortality decreased with an increase in maternal education. The pattern suggests that children of educated mothers experienced lower mortality rates than children of uneducated mothers. However, the wide overlapping confidence intervals suggest that the observed differences may merely reflect sampling variability rather than the true differences in the population.

Figure I: Child Mortality by Maternal Education

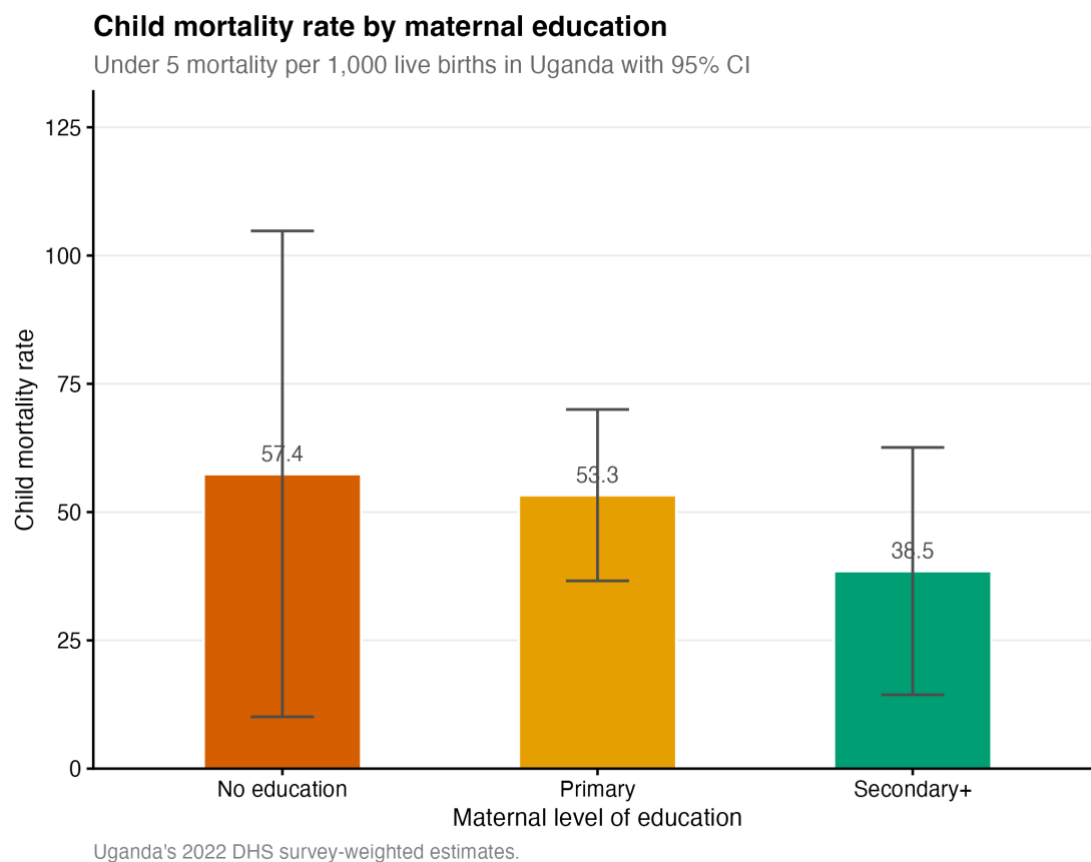


Figure II reveals higher mortality rate among children residing in rural households than urban areas. The wide and overlapping confidence intervals, however, suggests that the estimates are imprecise, and possibly, no substantial difference between rural and urban child survival outcomes.

Figure III depicts a U-shaped relationship between birth order and mortality. The first-born children experienced higher mortality rate than second- and third- born children.

Conversely, second and third born children experienced lower mortality rates compared to fourth born children. These differences, however, are imprecise as shown by the wide overlapping confidence intervals.

Figure II: Child Mortality by Place of Residence

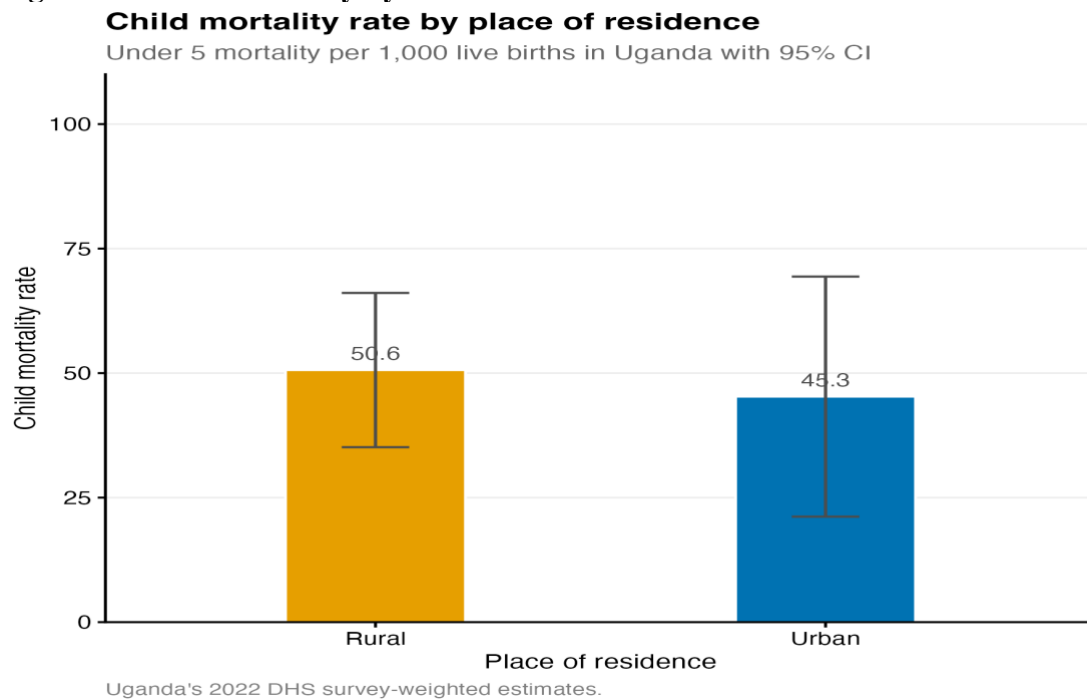


Figure III: Child Mortality Rate by Birth Order

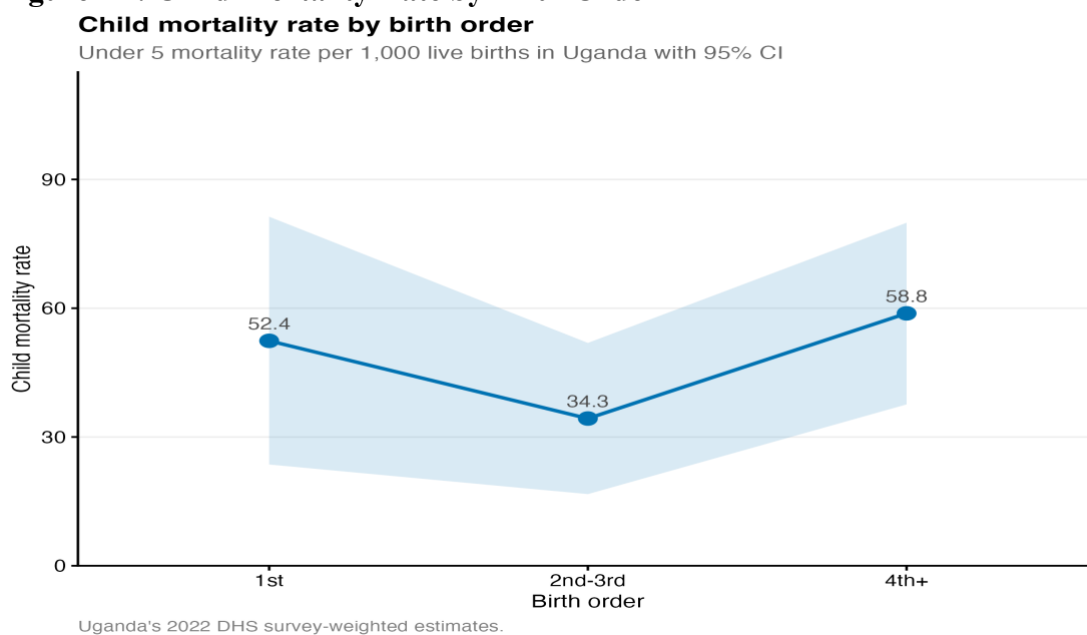
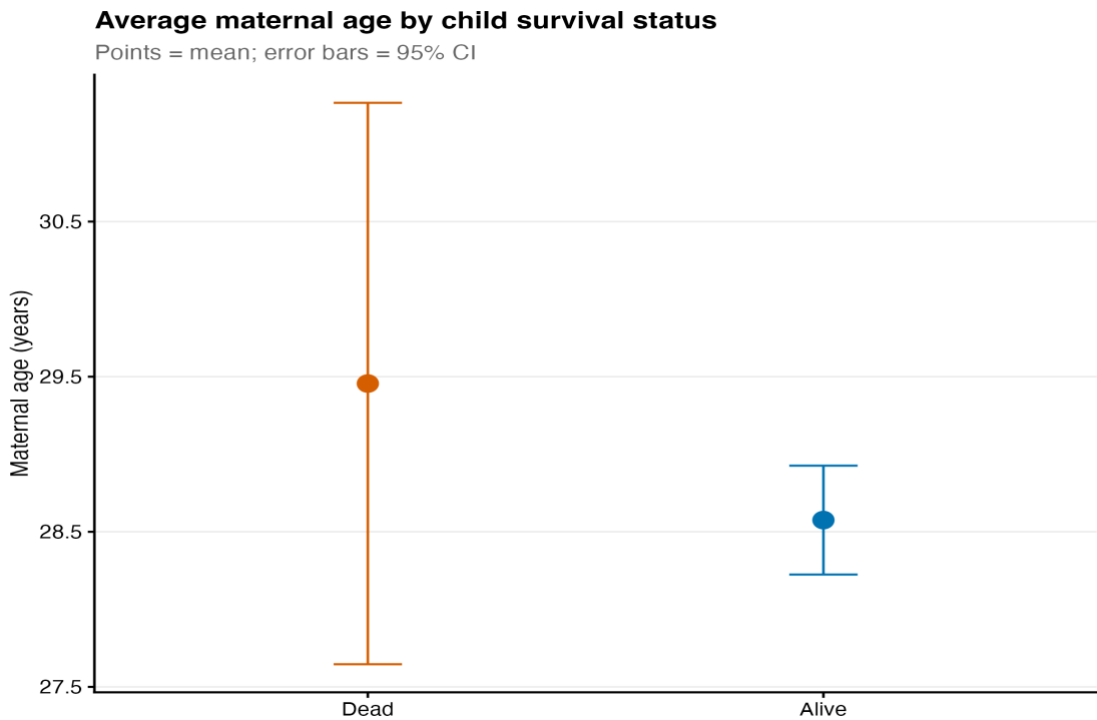


Figure IV illustrates that mothers of deceased children were slightly older than mothers of surviving children. However, the difference in mean age was small and insignificant. In addition, the confidence interval for deceased children is extremely wide, suggesting an imprecise estimate. Though the confidence interval for surviving children reveals a precise estimate, it overlaps with that of dead children. Hence, there was no difference in child survival by maternal age.

Figure IV: Average Maternal Age by Child Survival Status



3.4. Multivariate Analysis

Table VII presents survey-weighted logistic regression estimates of the determinants of child mortality. Model 1 includes socioeconomic factors, Model 2 adds demographic factors, and Model 3 adjusts for environmental factors. The interpretations below focuses on the full model (Model 3).

Demographic factors showed stronger and significant association with child mortality. Compared with children born within an interval shorter than 24 months, children born after an interval of 24 to 47 months had 45% lower odds of mortality (AOR = 0.55). A child born after 48 months had a 20% lower odds of mortality (AOR = 0.80) compared

to one born within 24 months. Though these associations are marginally significant, they suggest that proper birth spacing actually improves child survival outcomes.

The total number of children ever born had a positive and significant association with child mortality. Each additional child born increased the odds of child mortality by 24% (AOR = 1.24). This result suggests that higher fertility increases the probability of a child dying before reaching five years. This is plausible for poor rural children because larger family sizes are likely to be burdened by health and nutrition costs.

The relationship between maternal education and child mortality was insignificant. The estimates in Model 3 reveal that having primary education is associated with 6% lower odds of mortality (AOR = 0.94) compared to having no level of education. In addition, children born to mothers with a secondary or higher level of education had 19% lower odds of mortality (AOR = 0.81) compared to those born to uneducated mothers. The confidence intervals implies that these results are statistically insignificant.

The effect of household wealth on child survival was also insignificant. The estimate in Model 3 suggests that children belonging to the richest households had 30% lower odds of mortality (AOR = 0.70) compared to those from poorest households. Other estimates are surprising. For example, the estimate for richer households, AOR = 1.32, indicates that children from richer households had 32% higher odds of mortality. As for poorer households, the odds of mortality was 25% compared to poorest households.

A households place of residence showed statistically insignificant association with child mortality. Relative to rural households, children residing in urban households had 8% higher odds of mortality (AOR = 1.08). But this is insignificant.

The relationship between water access and child mortality was not significant, and it contradicts theoretical expectations. The coefficient suggests that children exposed to improved water sources had 75% higher odds of mortality (AOR = 1.75) compared to those exposed to unimproved water sources. The relationship is, however, insignificant. Given that the majority of the households used borehole and communal water sources, this result reflects unobserved factors such as water quality, the quality of water storage

and fetching equipment, and overall sanitation. Thus, having access to a water source should not necessarily be interpreted as access to safe water.

Overall, the multivariate results reveal demographic factors such as the total number of children ever born and birth spacing as the main predictors of child mortality in Uganda. Though socioeconomic and environmental factors were insignificant determinants, they still interestingly indicates that maternal education and access to safe water are vital for the survival of a child in Uganda.

4. POLICY IMPLICATIONS AND LIMITATIONS

4.1. Policy Implications

The main predictors of child mortality in Uganda were found out to be the total number of children ever born and birth spacing. These results imply that larger family sizes and short birth intervals increase the probability of a child dying before reaching five years. Thus, reducing or eradicating child mortality in Uganda requires a stern policy focus on family planning, reproductive healthcare, and access to child healthcare. Interventions that promote optimal birth spacing and minimal number of children, especially among poor rural households, are required to reduce child mortality to less than 25 deaths per 1,000 live births by 2030. In addition, it is also important to increase access to quality education because uneducated mothers experienced higher mortality rates. Also, access to water sources is not enough. Effort should be put into improving the safety of existing water and overall sanitation of equipment used for fetching and storing water.

4.2. Limitations

This analysis faced some limitations that demand cautious interpretation of the findings. The dataset used was a cross-sectional survey data. This only allowed for correlational identification of effects. In addition, some potentially relevant determinants were not analyzed, for example, access to antenatal care, paternal education, and the quality of healthcare services. Future research could include these factors to see if this paper's results remain the same. However, by using a representative sample, this study provide a nationally representative evidence on the determinants of child mortality in Uganda.

Table VII: Survey-weighted logistic regression analysis of child mortality determinants

	Model 1	Model 2	Model 3
	Socioeconomic	+ Demographic	+ Environmental
	AOR (95% CI)	AOR (95% CI)	AOR (95% CI)
Mother's Education Level (Ref: No education)			
Primary	0.95 (0.37 – 2.40)	0.95 (0.38 – 2.40)	0.94 (0.37 – 2.34)
Secondary+	0.72 (0.23 – 2.25)	0.86 (0.25 – 2.88)	0.81 (0.24 – 2.70)
Household Wealth Index (Ref: Poorest)			
Poorer	0.73 (0.30 – 1.78)	0.74 (0.31 – 1.79)	0.75 (0.31 – 1.80)
Middle	1.01 (0.46 – 2.22)	0.97 (0.44 – 2.17)	1.01 (0.45 – 2.28)
Richer	1.28 (0.60 – 2.76)	1.30 (0.59 – 2.89)	1.32 (0.59 – 2.96)
Richest	0.67 (0.24 – 1.89)	0.72 (0.25 – 2.04)	0.70 (0.24 – 2.02)
Place of Residence (Ref: Rural)			
Urban	1.03 (0.51 – 2.08)	1.09 (0.53 – 2.26)	1.08 (0.51 – 2.25)
Birth Interval (Ref: < 24 months)			
24 – 47 months		0.54* (0.28 – 1.06)	0.55* (0.28 – 1.08)
48+ months		0.77 (0.37 – 1.62)	0.80 (0.38 – 1.69)
Total Children Ever Born			
		1.22** (1.03 – 1.45)	1.24** (1.04 – 1.48)
Mother's Age (Years)			
		0.97 (0.90 – 1.04)	0.96 (0.89 – 1.04)
Water Sources (Ref: Unimproved)			
Improved			1.75 (0.72 – 4.27)
Observations	1434	1434	1434

Notes: Survey-weighted logistic regression estimates using Uganda 2022 DHS sampling weights. AOR = adjusted odds ratio; CI = confidence interval; Ref. = Reference category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5. CONCLUSION

This analysis examined the determinants of child mortality in Uganda using data from the Uganda 2022 Demographic and Health Survey. Particularly, a sample size of 1,434

children obtained from the Kid's Record file was extracted, cleaned, and analysed using statistical software R and Stata. Throughout the analysis, DHS sampling weights were applied to produce descriptive and survey-weighted logistic analysis.

Overall, the estimated 2022 child mortality rate was 49 deaths per 1,000 live births. This was above the SDGs target of less than 25 deaths by 2030. The majority of the children lived in poor rural households, where the child mortality rate was highest. Most children lived in Buganda sub-region and in households with an average of four children. Almost half of the children were fourth or higher borns, born to adult mothers who were at least 28.7 years old. It took an average of 28.3 months to give birth to the next child, and most mothers had at least a primary and secondary education. The multivariate analysis reveal that birth spacing and total number of children were the main determinants of child mortality. Having many children and shorter birth intervals was associated with higher odds of child mortality in Uganda.

These findings imply that the government of Uganda and child health advocates should strengthen and or introduce policies and interventions that increase access to family planning and child healthcare. Particularly, these interventions should encourage optimal birth spacing of at least over 24 months after the preceding birth and a small number of children. Since the majority of households are rural and poor, these policies should target poor rural mothers if Uganda is to reduce child deaths to less than 25 by 2030.

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